Privacy-Preserving Shopping Intelligence via Federated and On-Device Learning

How much next-product recommendation quality survives when shopping data never leaves the device?

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MSc in Artificial Intelligence and Data Science
School of Computing, Queen's University · October 20, 2026



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The research question

“How much shopping recommendation quality can be retained when moving from cloud-based to on-device or federated models?”

Why it matters

- A browser sees shopping behaviour on the user's own device
- Cloud training copies that history to a server, which users and regulation resist
- Learning on the device is constrained: little compute, one short history, and updates can leak

What we measure

- **Retention** of cloud quality: federated and on-device systems
- **Server pretraining** (warm start) and **on-device fine-tuning**
- **Privacy**: cost of DP, leakage audits
- **Deployability**: INT8 size, browser latency, upload bytes

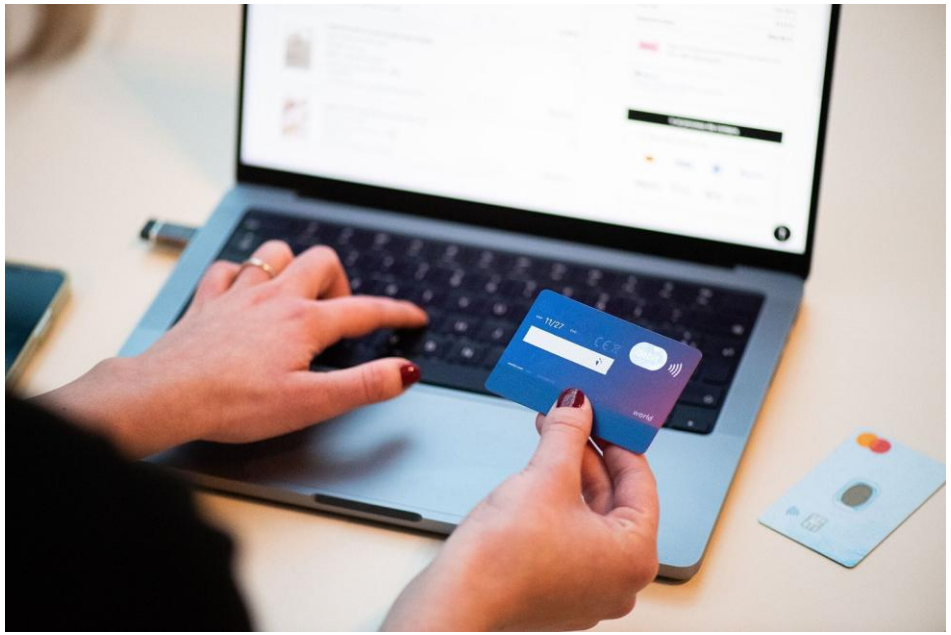
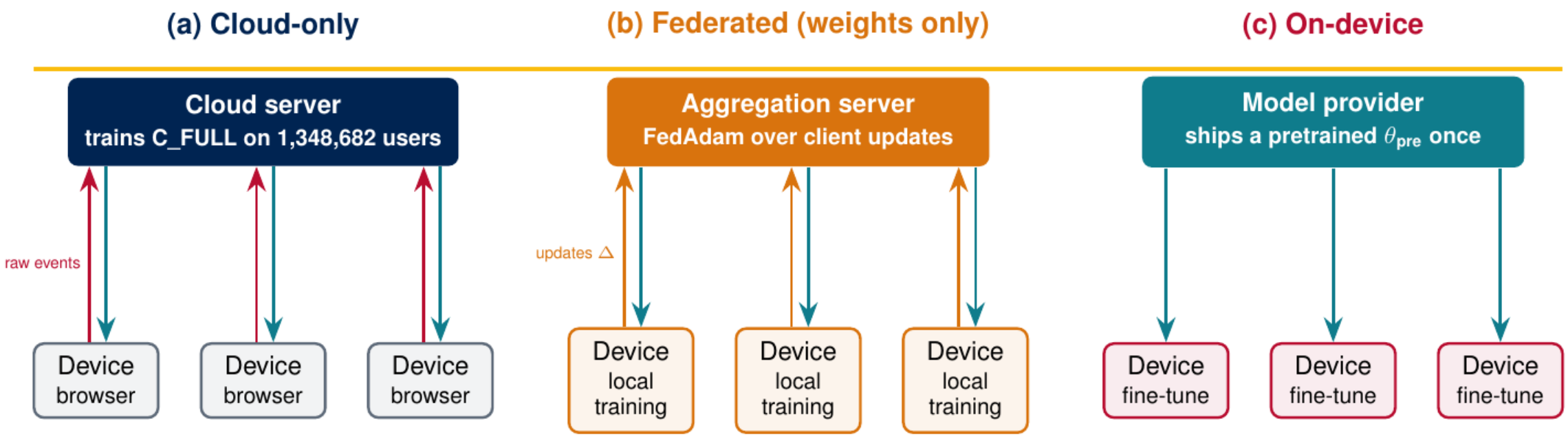


Photo: Bogdan Hoyaux / European Commission, CC BY 4.0, Wikimedia Commons

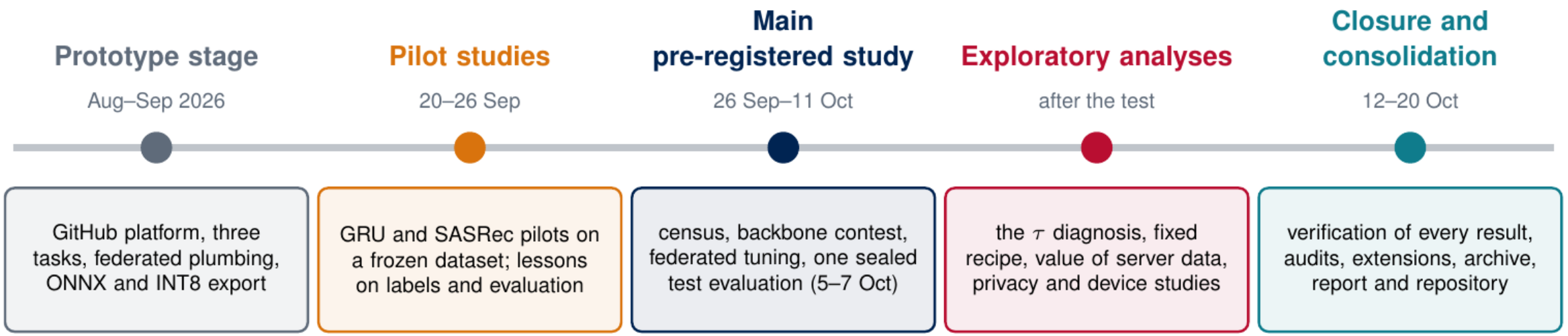
Three ways to build a shopping recommender

Headline results are retention: the share of cloud-model quality kept



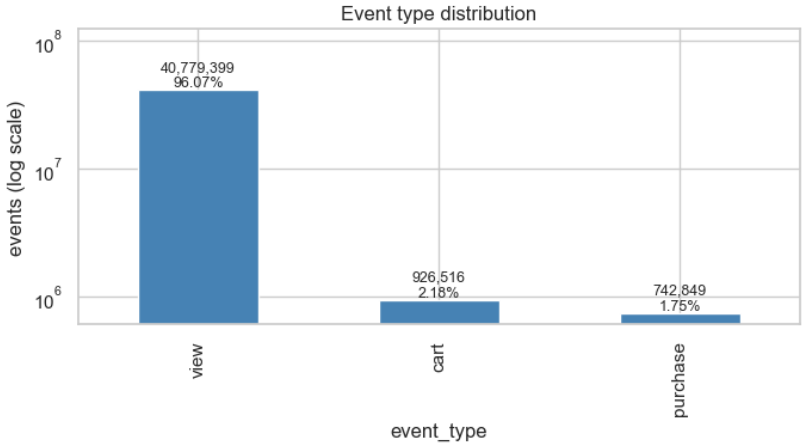
The project at a glance

Each stage answered a question raised by the previous one

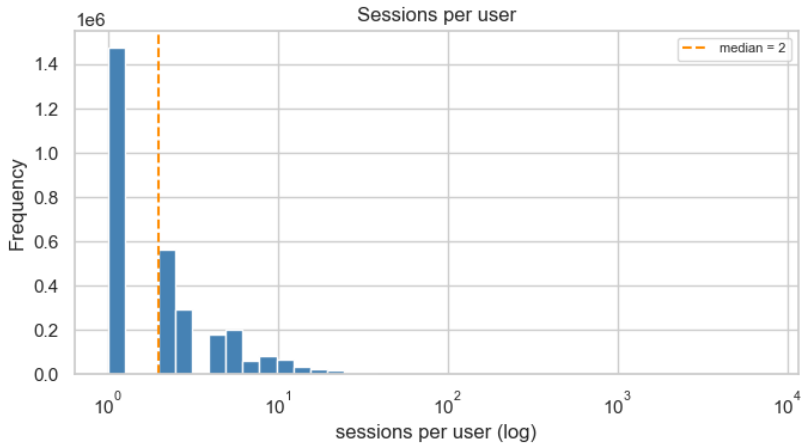


The data: real shopping sessions from one store

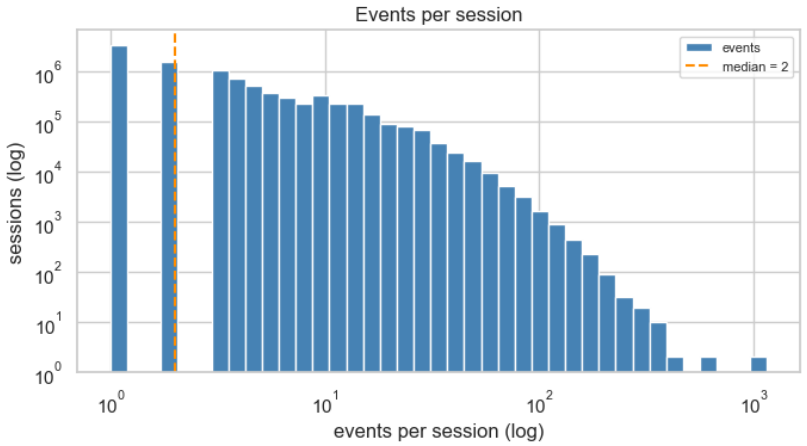
REES46, October 2019 (data-audit notebook); the task uses views only



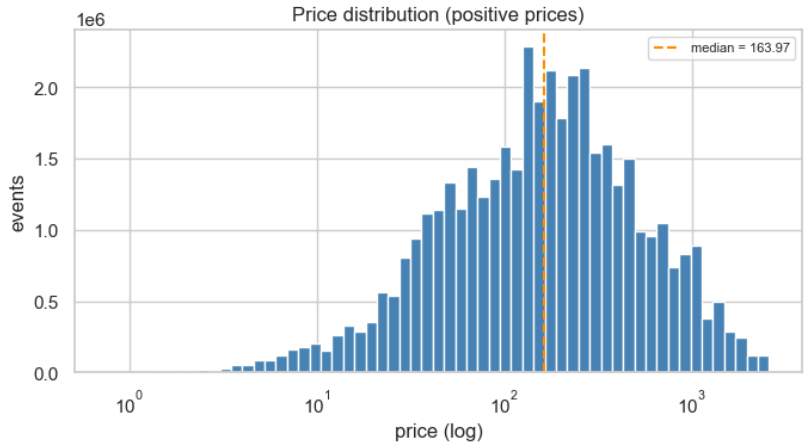
Event types: 96.07% views, 2.18% carts, 1.75% purchases



Sessions per user: median 2



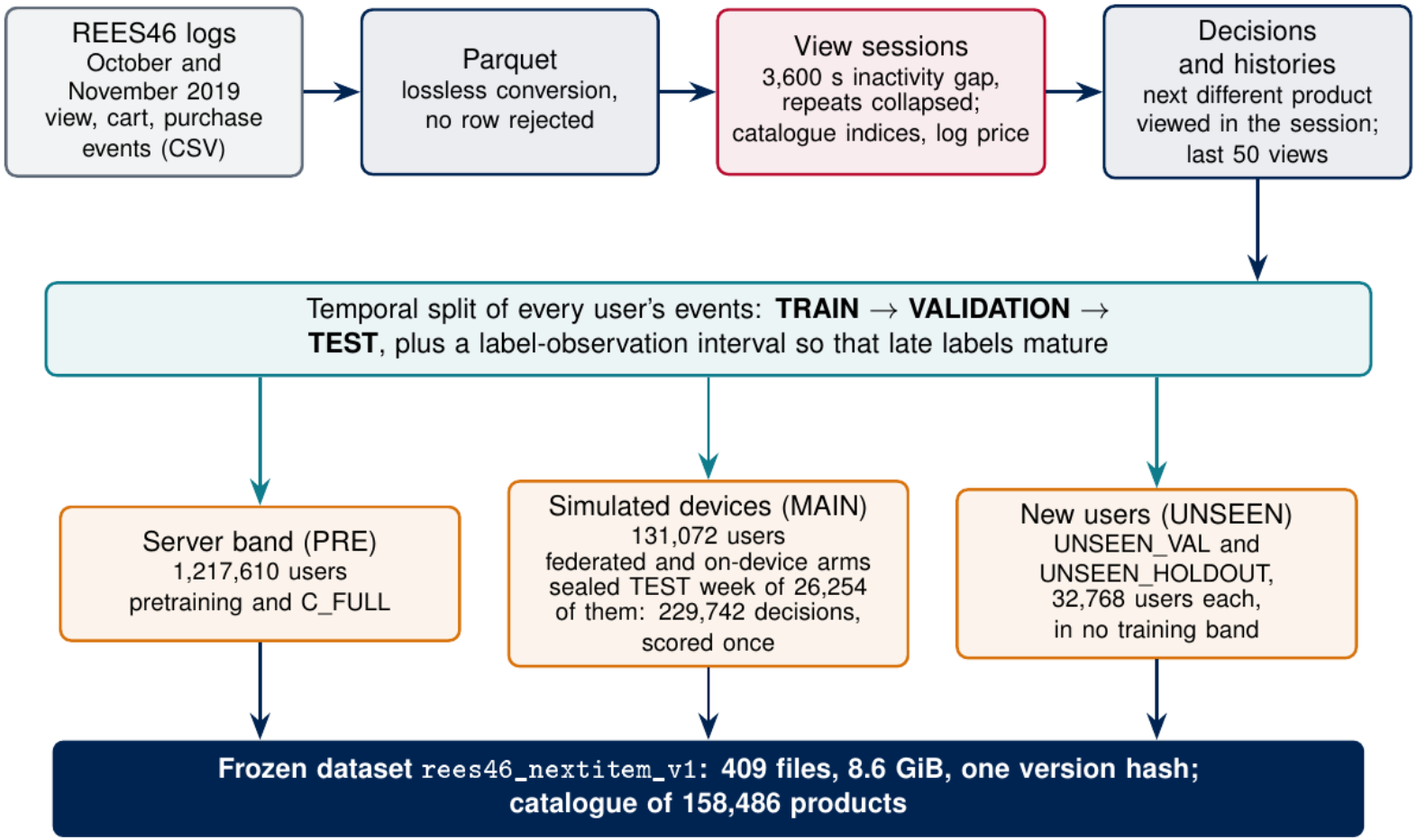
Events per session: median 2, long tail



Product prices: median 163.97

From the raw log to the frozen dataset

Oct–Nov 2019 · full catalogue of 158,486 products · one frozen dataset with a version hash

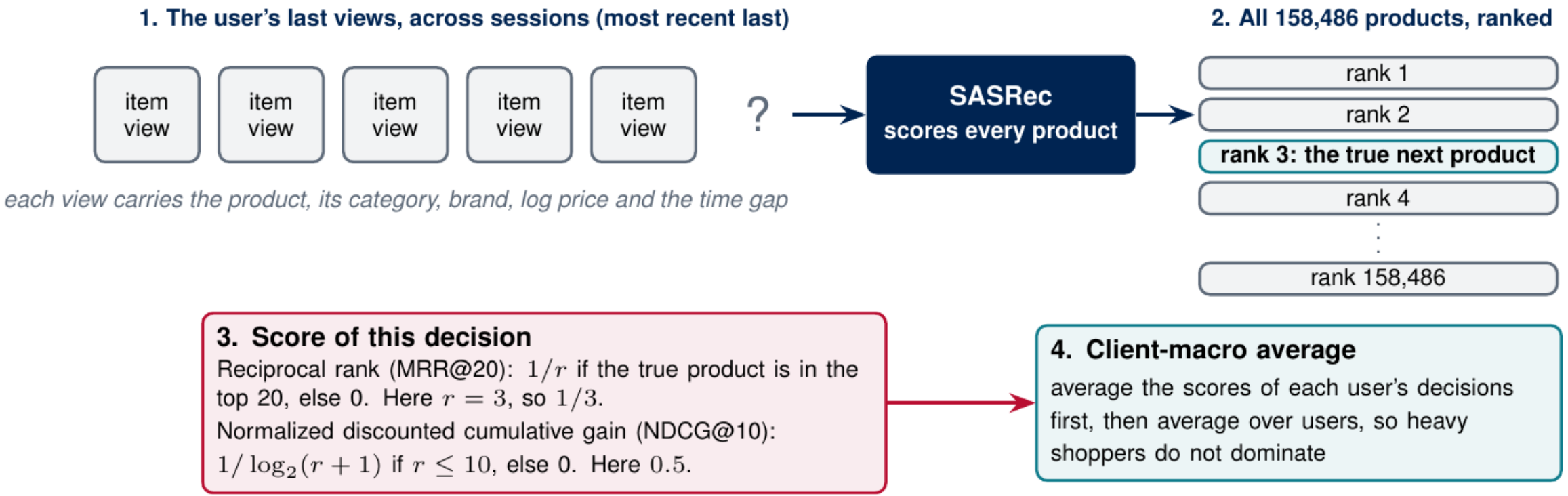


Design choices

- Rank the **full catalogue**, no sampled negatives
- Disjoint user bands; sealed TEST is the devices' last week
- SASRec: device d=64 (20.7M) · cloud d=256 (83.9M)
- Every decision locked before **one** TEST look

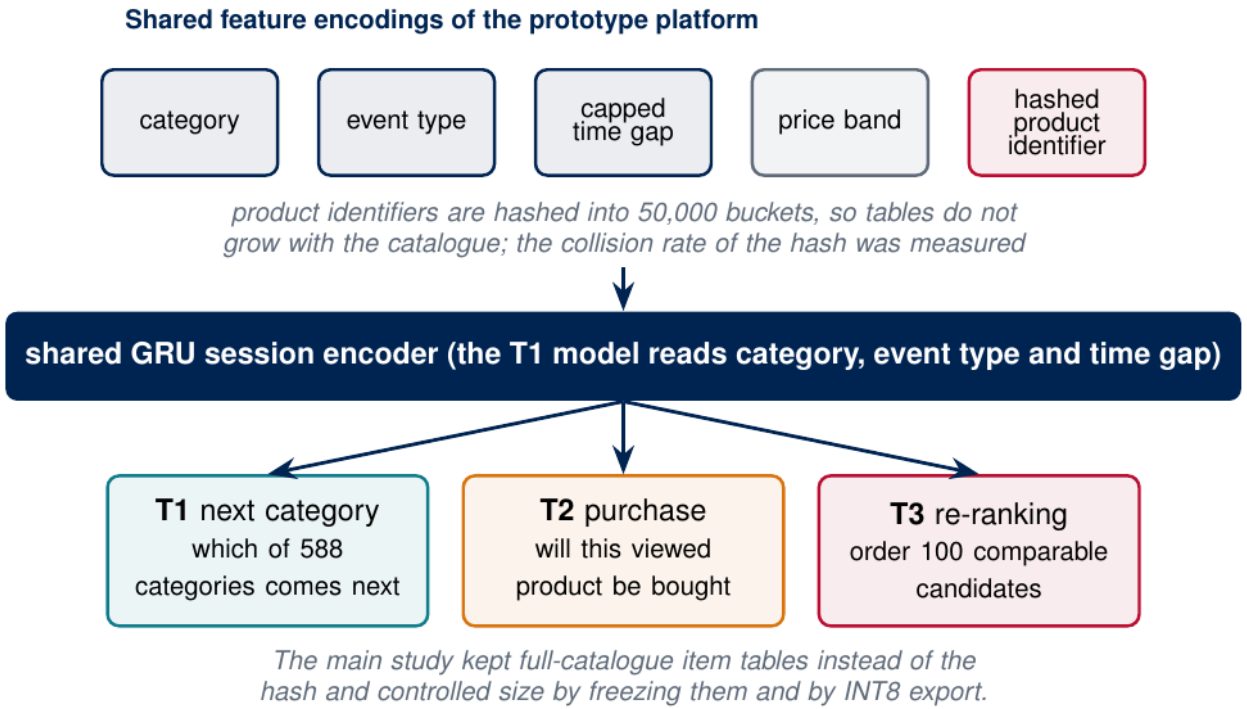
How one recommendation is scored

Mean reciprocal rank (MRR@20), averaged per user, then over users



The prototype platform

August–September 2026, on GitHub

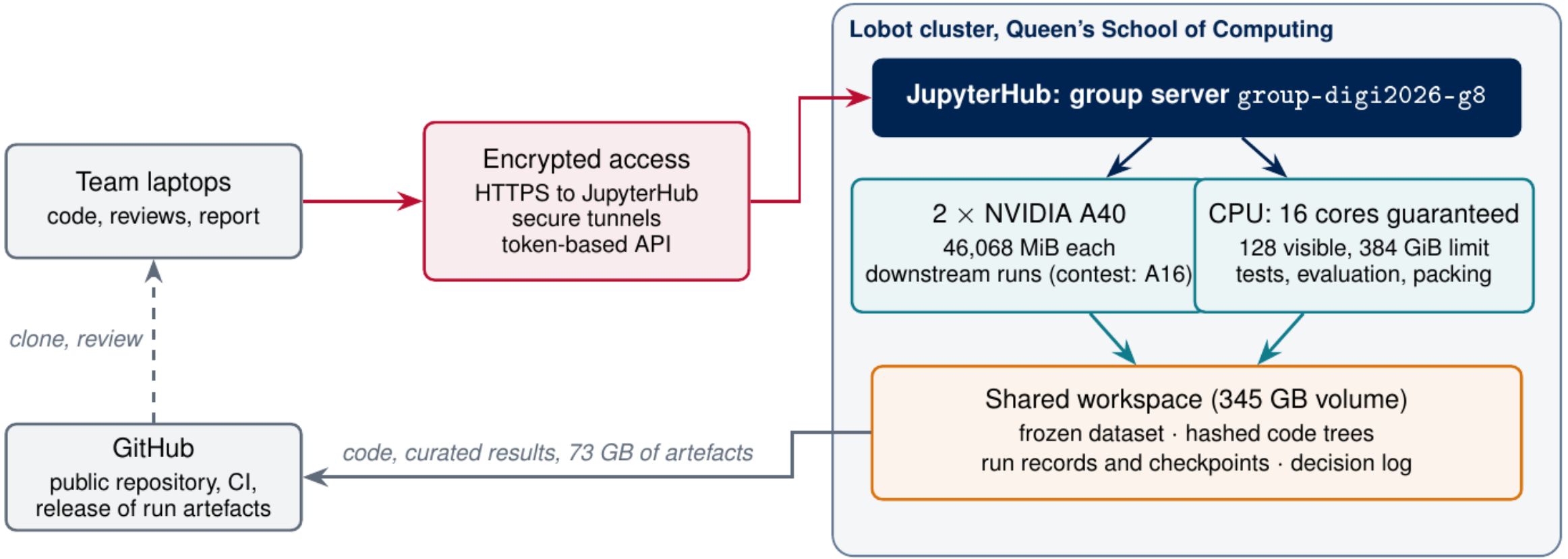


What the prototype delivered

- Data pipeline: 42.4M events → canonical events, sealed TEST window
- Three tasks; on T1 the GRU beat the transition-table baseline
- Federated plumbing with Flower; T1 pilot retention 88.38% (validation)
  - Weak matched pilot reference: not comparable with the main study
- Hashed identifiers, price bands, ONNX and INT8 export, byte ledger, CI
- Lesson: a label defect and evaluator conventions → a stronger protocol

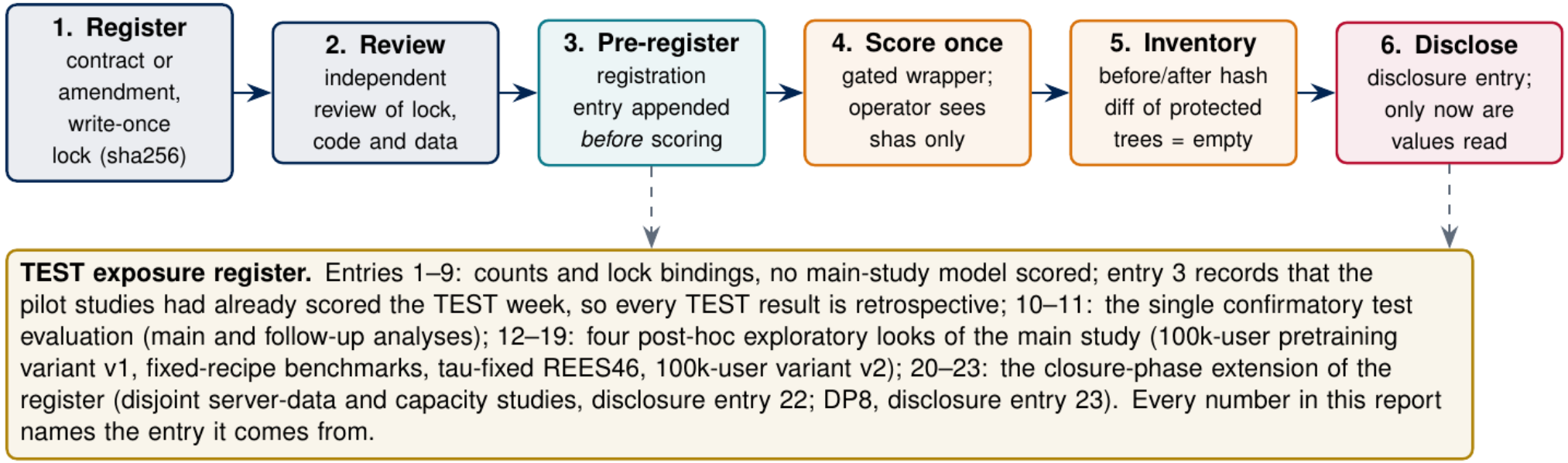
Where the study ran

Queen's Lobot cluster: 2× A16 for pilots and contest, then 2× A40



The pre-registered main study

After the pilot studies: a backbone contest, then one sealed test evaluation



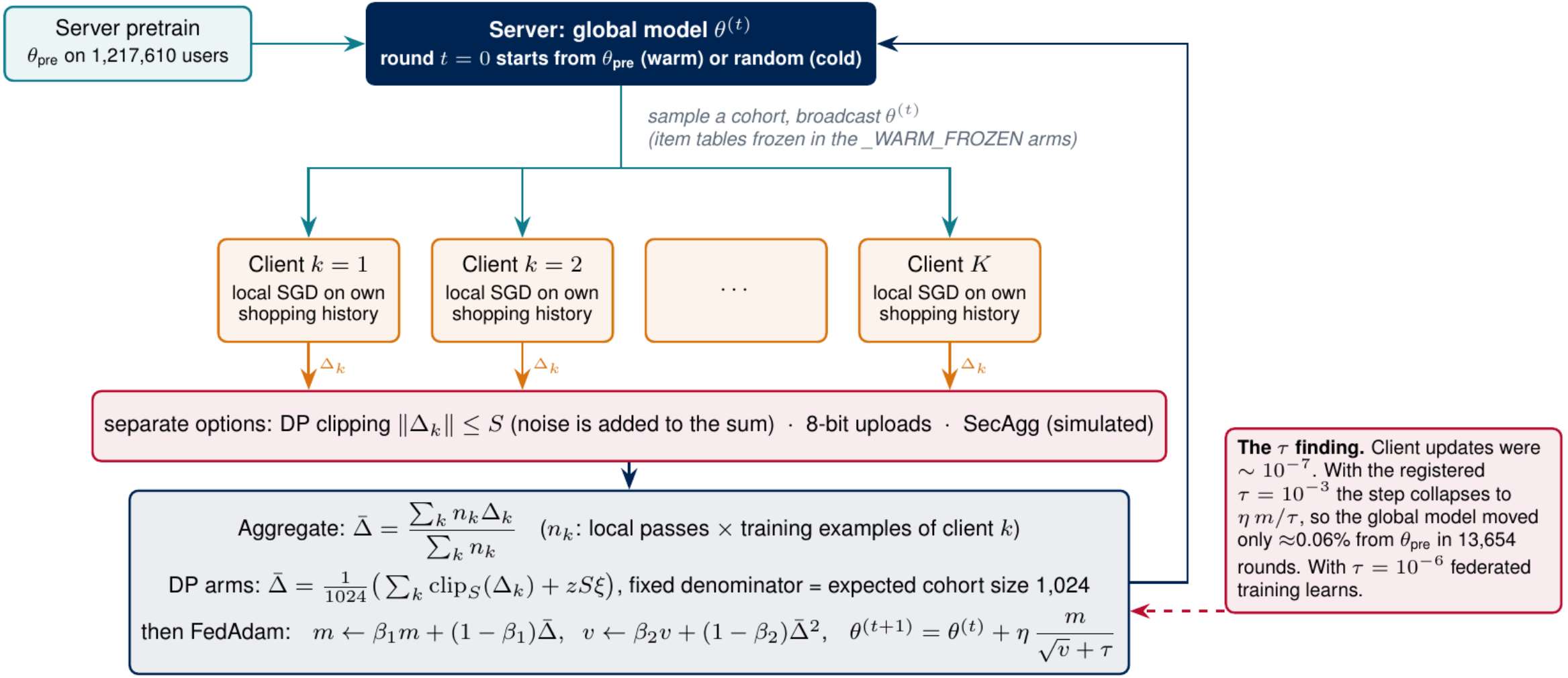
Backbone contest (validation)

- GRU family mean 0.1569
- SASRec family mean 0.1612
- SASRec locked as the backbone

Why so much protocol?

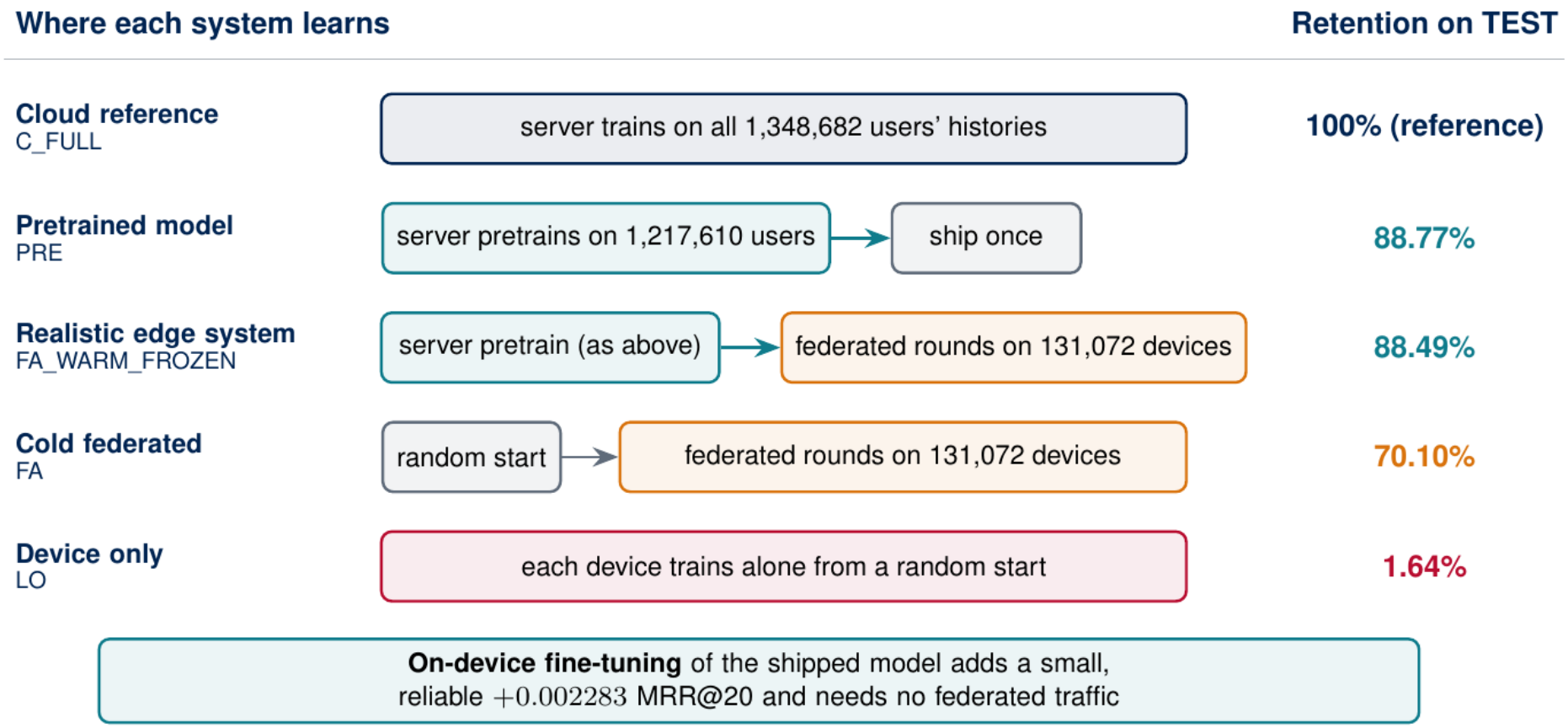
- 33 main-analysis contrasts, two seeds, many arms: easy to fool ourselves
- Selection only on calibration data; validation read once; locked methods scored on TEST once (retrospective: the pilots had seen that week)
- Every later exploratory test look is a numbered, disclosed register entry

The federated recipe: warm-started FedAdam



The confirmatory answer

REES46 TEST, 26,254 users, client-macro MRR@20; devices are simulated *(confirmatory result)*

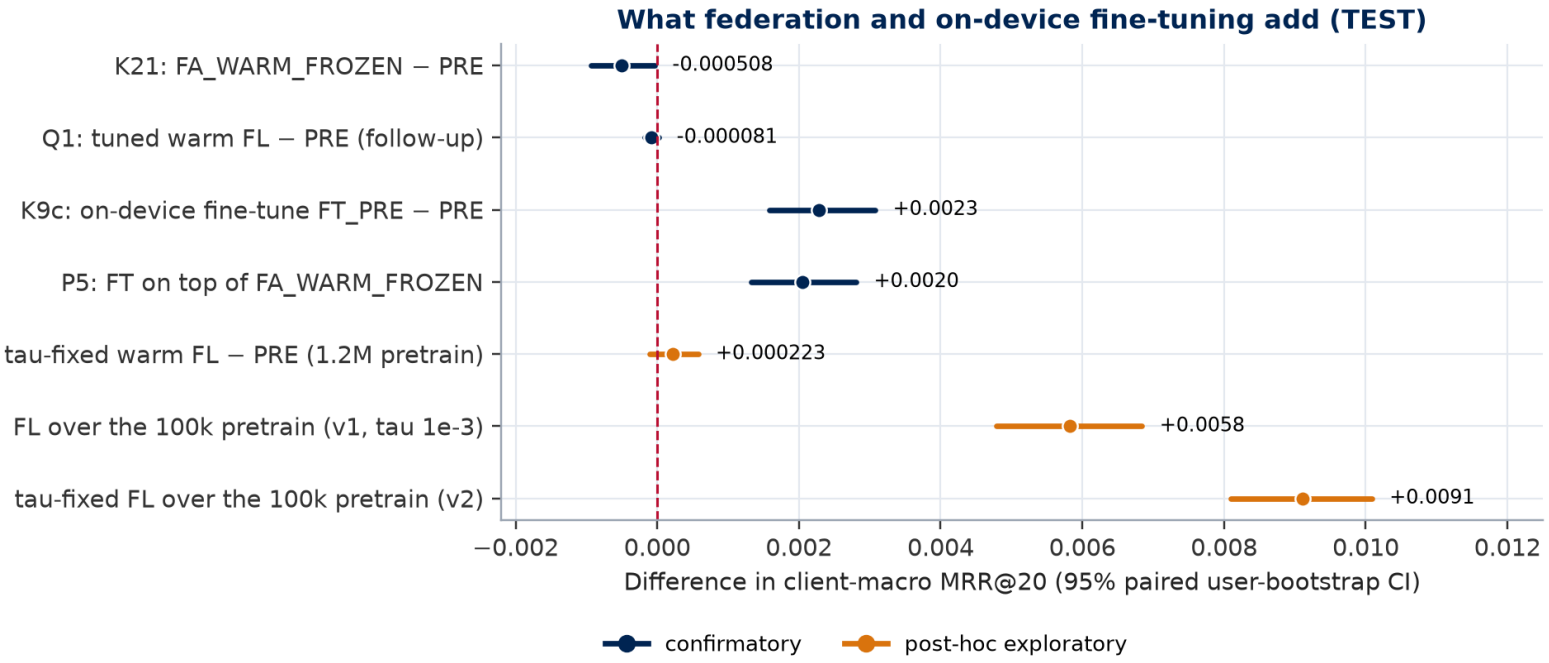


Retention: client-macro MRR@20 of the system as a percentage of the cloud reference, on the sealed TEST week of MAIN users (confirmatory).

FA and LO vs C_BIG (same users): 85.11% and 2.10%

What federation and fine-tuning add

Differences in MRR@20 with 95% paired-bootstrap intervals

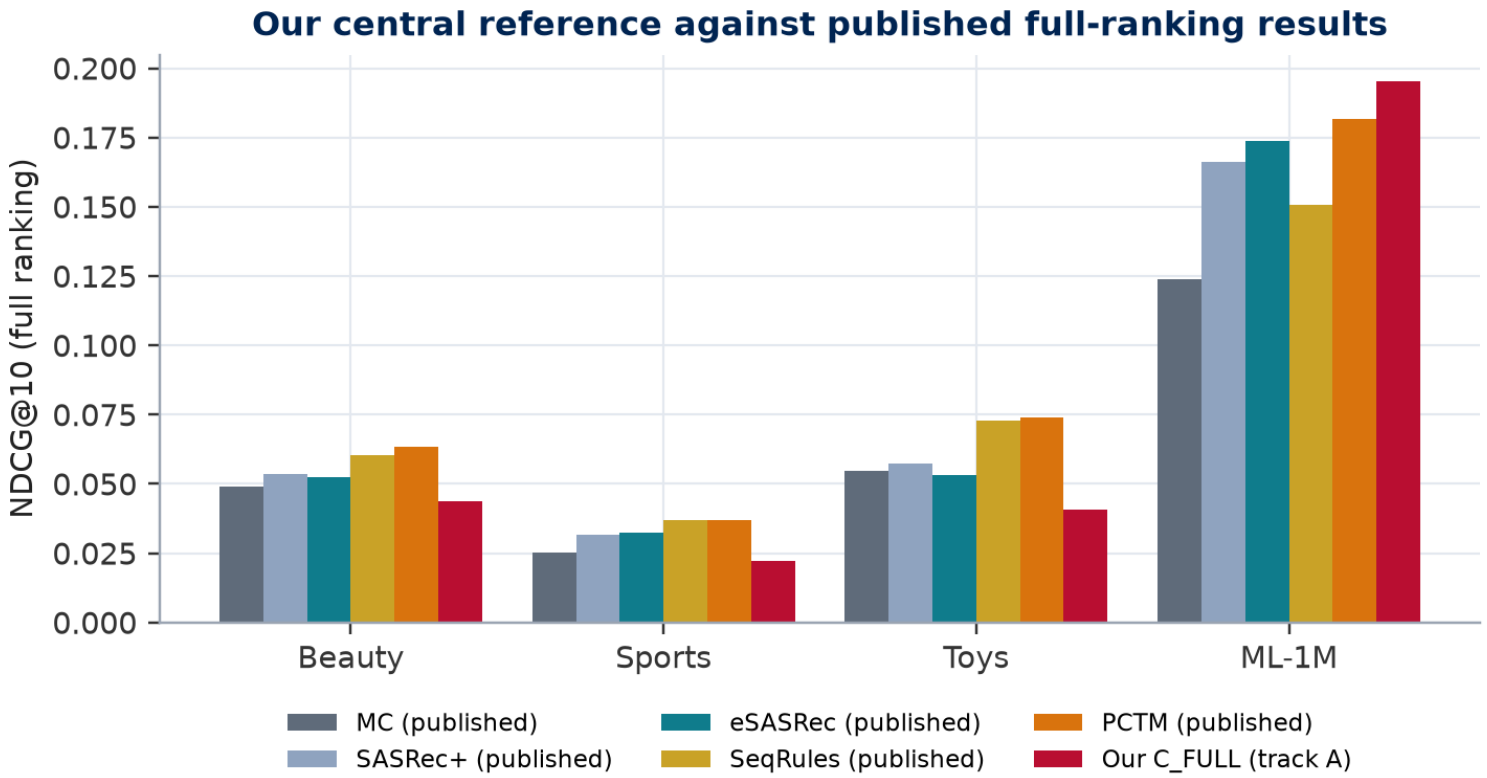


Reading

- **K21** FL – PRE: −0.000508 [−0.000933, −0.000037]: slightly worse than the pretrain alone
- **Q1** tuned warm FL – PRE: equivalent (±0.0005)
- **K9c** on-device fine-tune: +0.002283, the deployable gain
- Orange: post-hoc looks (next slides)

Benchmarks: our cloud reference against published results

Full-catalogue NDCG@10 on public datasets

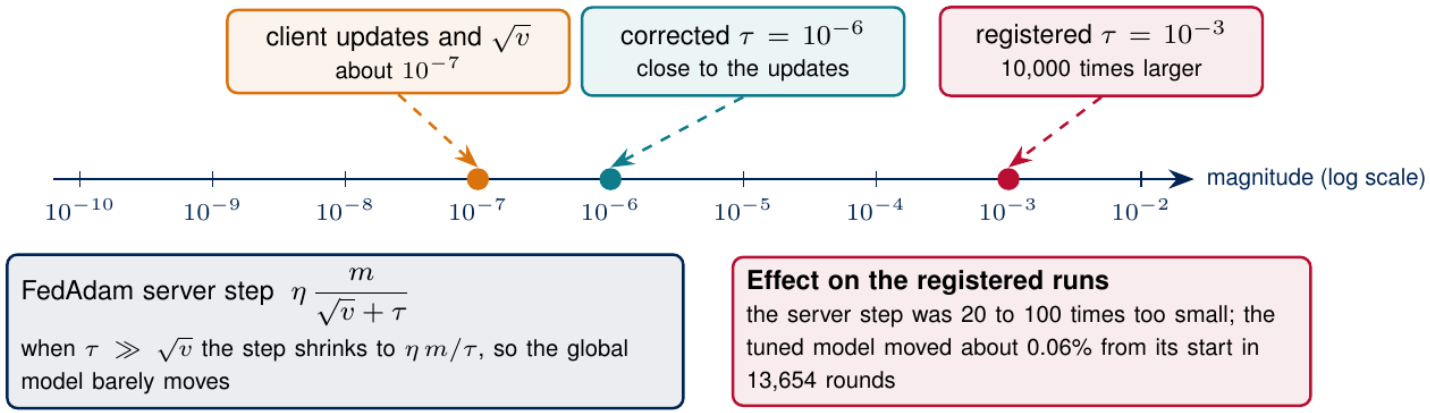


Confirmatory benchmark looks

- Credible, not state of the art: below SeqRules and PCTM on Amazon
- MovieLens-1M: 0.1936 against 0.1739 published for eSASRec
- Best federated arm kept only 13–77% of it
- → something in the federated recipe was wrong

Diagnosis: FedAdam's τ strongly suppressed federated training

Found after the test evaluation *(exploratory result)*



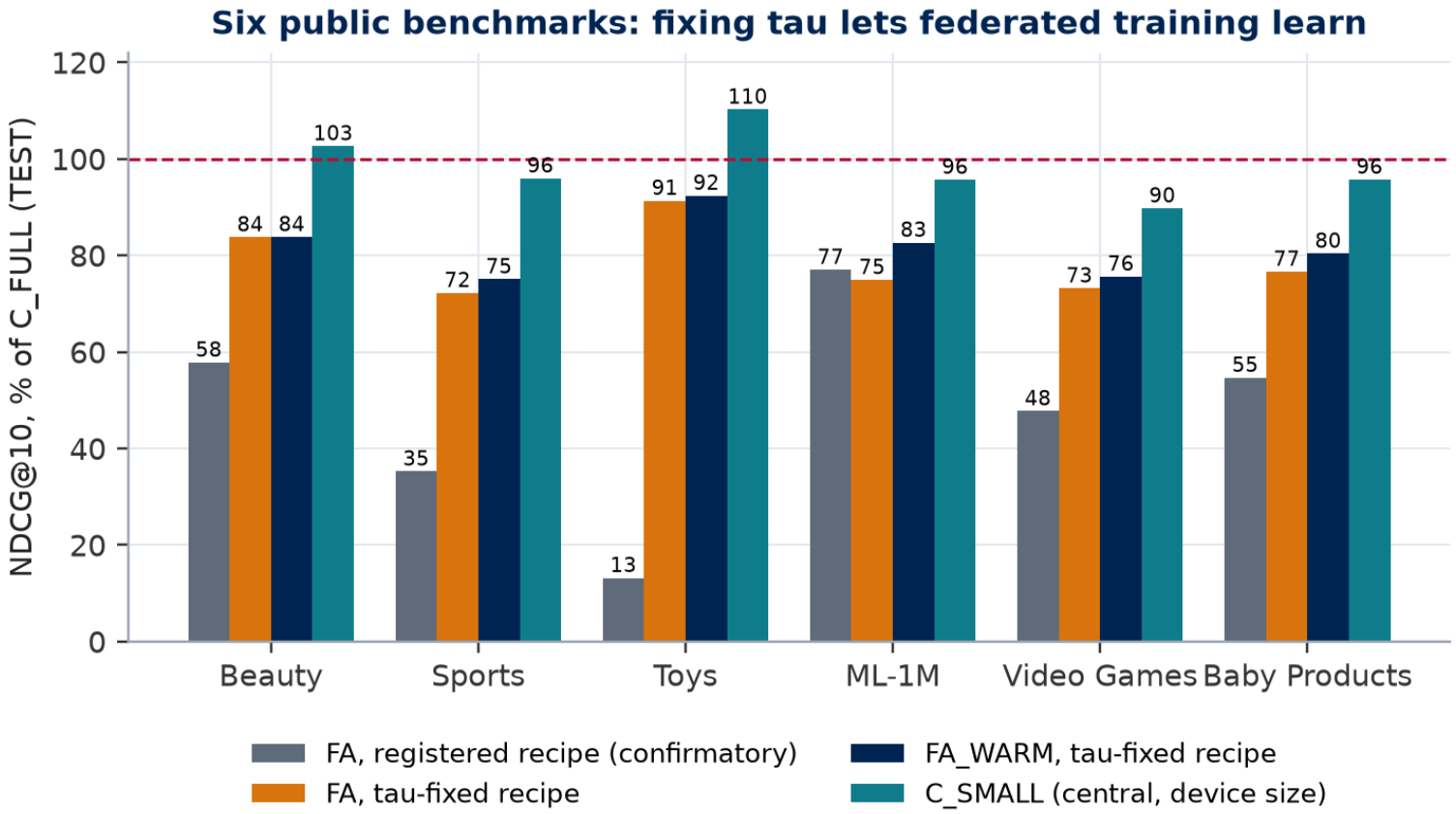
Mechanism

- FedAdam step ∝ m / (√v + τ)
- Updates ≈ 10^{-7}, registered τ = 10^{-3}
- → the server step was 20–100× too small
- The tuned model moved ≈ 0.06% in 13,654 rounds
- Not the only cause: FedAvg cells without τ also matched PRE
- Confirmatory results stay as registered; the fix is exploratory

With a corrected recipe, federated training learns

Six public benchmarks, TEST (exploratory look)

(exploratory result)

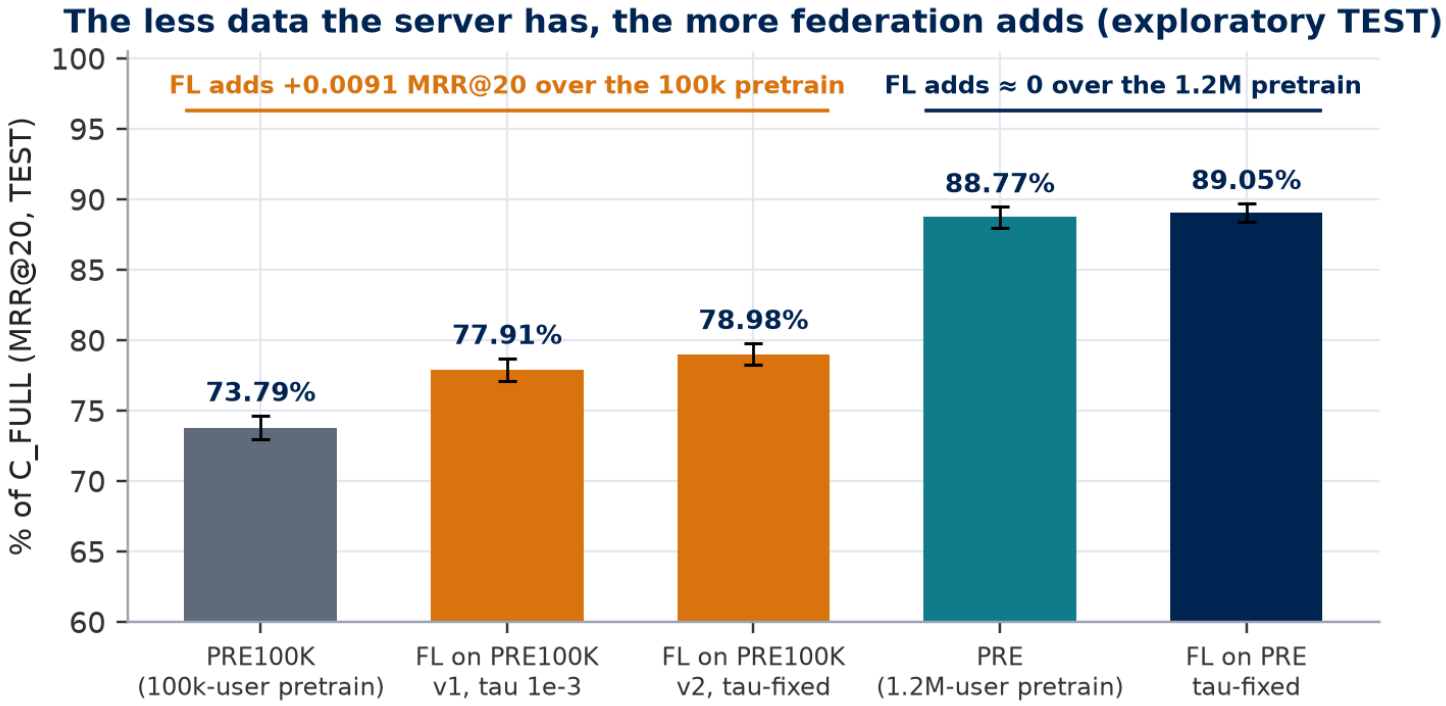


What changed

- Best federated arm: 13–77% → 75–92% of the cloud model
- Device-size central model 90–110%: size mostly not the bottleneck
- REES46: warm FL still adds nothing measurable (+0.00022, CI includes 0)
- Caveat: τ, server step, budget and item tables changed together

Federation added more when the server knew less

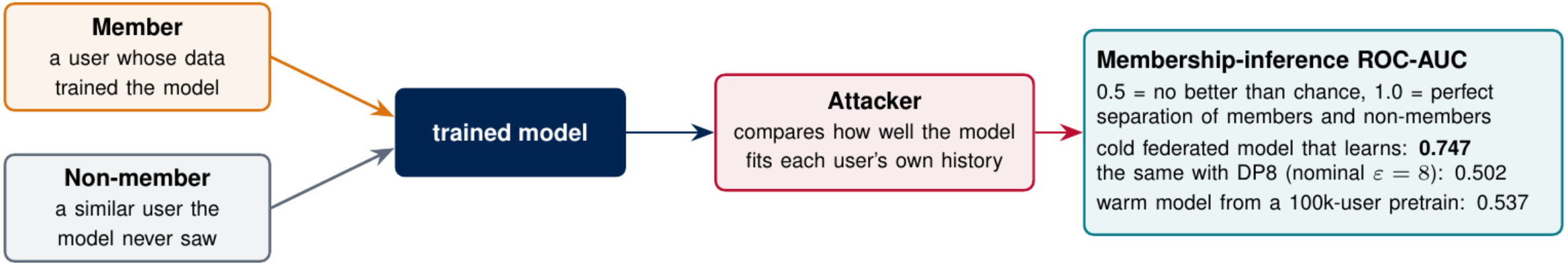
(exploratory result)



Evidence

- REES46, 100k-user pretrain: federation adds **+0.0091** MRR@20
- REES46, 1.2M-user pretrain: federation adds ≈ 0 (+0.00022)
- Benchmarks (NDCG@10, TEST): bigger server band, bigger warm-start gain
- Federation does not replace server data: 78.98% vs 88.77%

Privacy: learning leaks, and guarantees need care



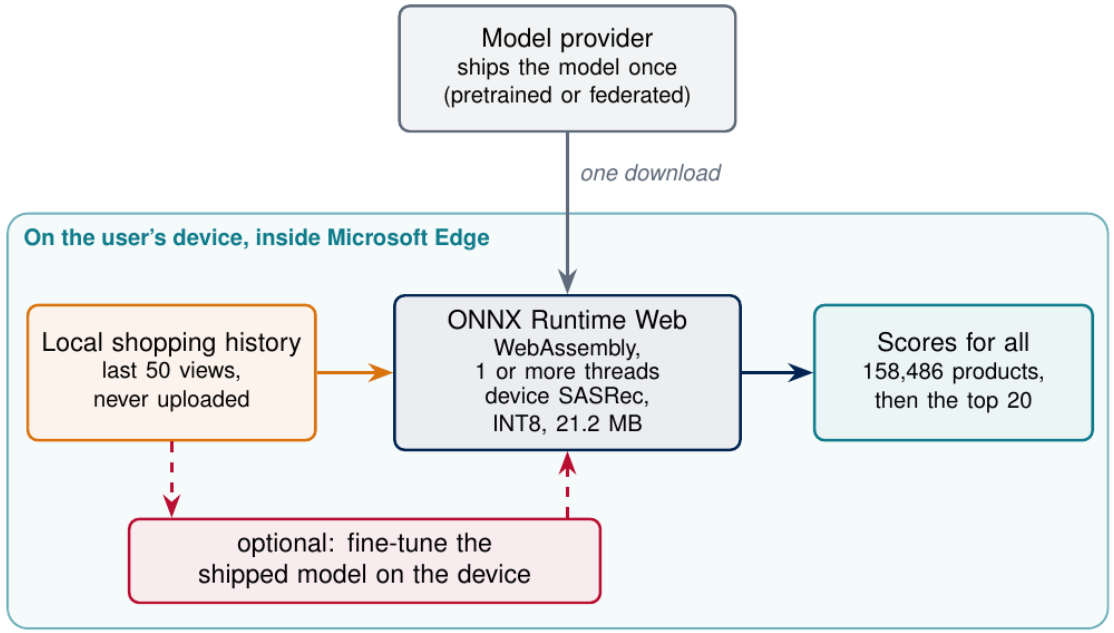
A model that fits its training users much better than unseen users gives membership away; differential privacy limits how much any one user can change the model.

Findings

- The cold federated model that learns leaks membership (AUC 0.747)
- Nominal ε = 8 brought this attack to chance (AUC 0.502)
- Whole DP8 recipe kept 94.71% (100k-user warm start), 31.30% (cold)
- Legacy DP: no formal guarantee; a passed audit is not proof
- Secure aggregation simulated; aggregate update attack 2.6 pp, above 2 pp limit

On-device feasibility in Microsoft Edge

Inference timed on laptop, phone and server CPU; fine-tuning simulated offline



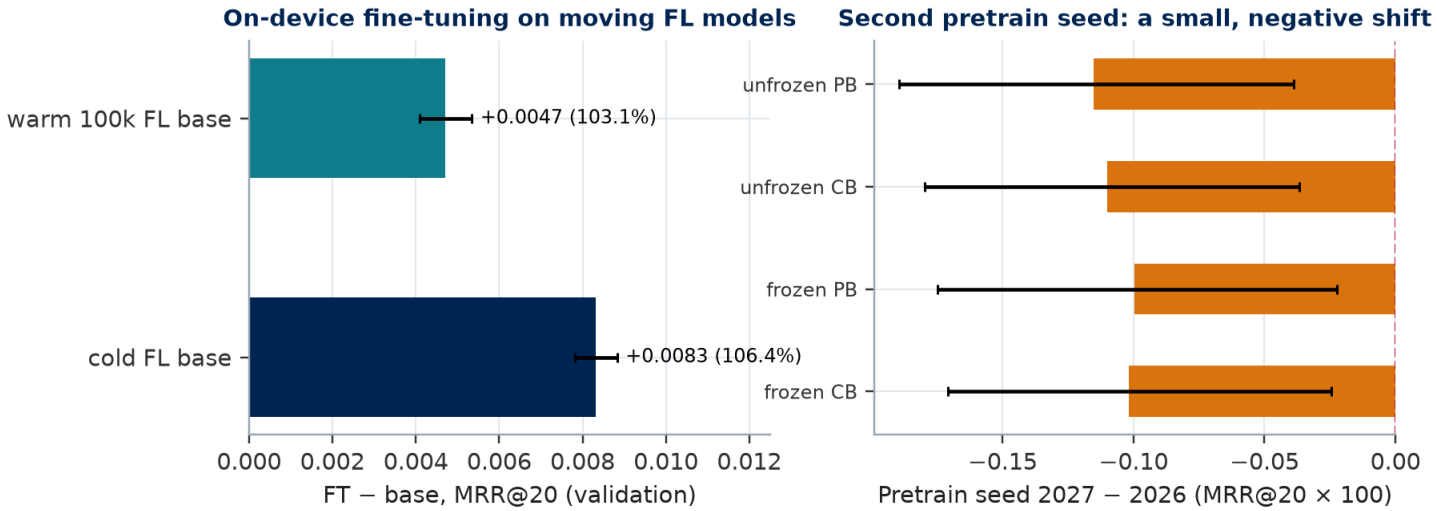
warm model call: 1.7–5.1 ms
cold start: 0.7–5.2 s
(median over runs, FP32 or INT8; inputs built before timing; top-20 selection not timed)

Deployability

- Tuned warm model: 82.9 MB FP32 → 21.2 MB INT8
- INT8 quality: 99.98% of FP32 (C_SMALL, 16,384 users)
- 8-bit uploads: 100.11% (cold line), 99.85% (warm line)
- Warm model call: 1.7–5.1 ms median (top-20 not timed)
- Phone cold start ≈ 2.5 s INT8, ≈ 5 s FP32
- INT8 not faster warm on phone or multi-thread server

Closure experiments

Technical closure and extensions, 12–17 October

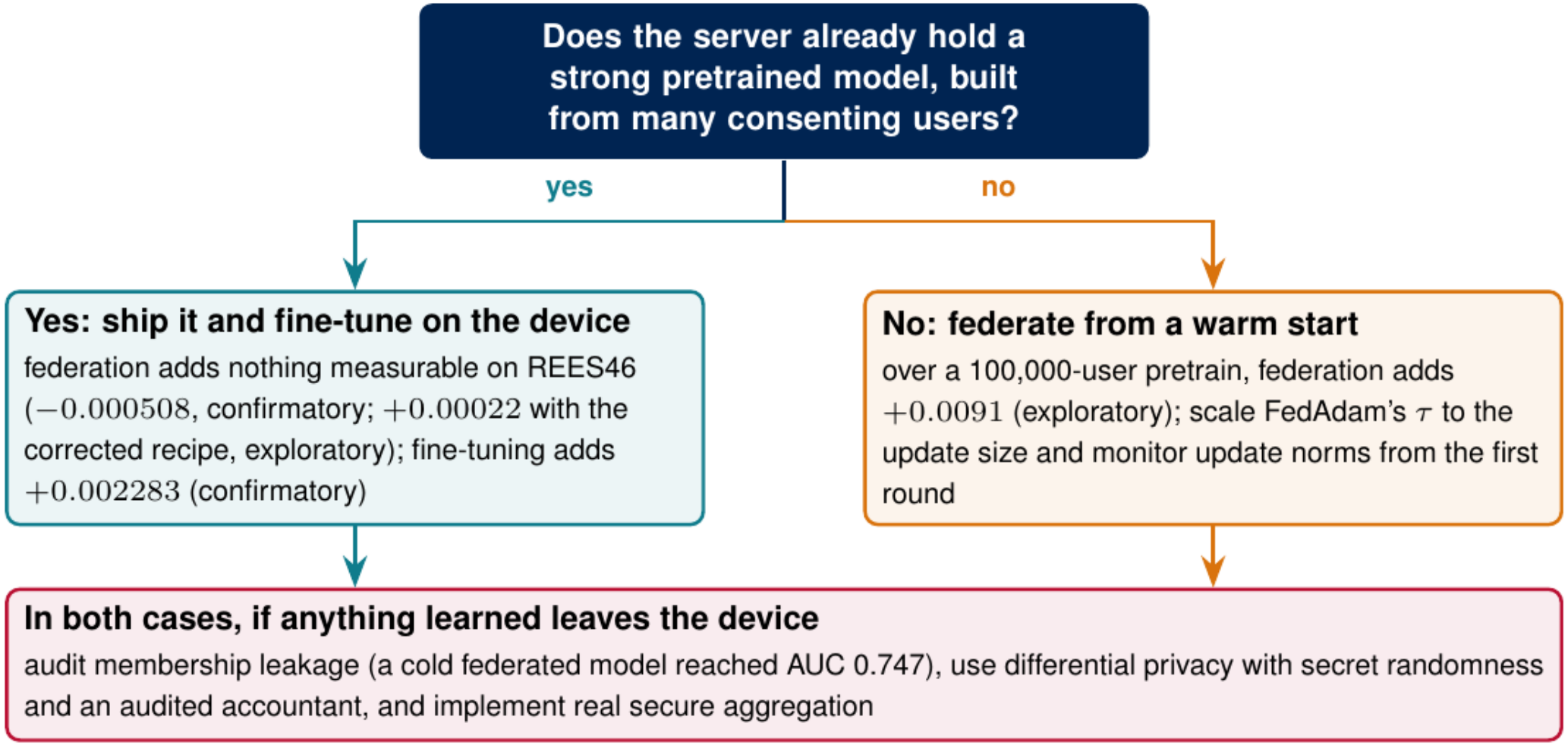


What closure added

- Second pretrain seed: small negative shift (−0.10 pts)
- Fine-tuning moving FL models: +0.0047 to +0.0083 MRR@20 (+3–6%, validation)
- Training-free baselines: best Markov chain 0.1366 (validation)
- Stopped or not run: DP-FTRL, canary restart, December replication
- All results labelled; no confirmatory result changed or re-scored

What this means for a browser shopping assistant

A heuristic from one main dataset (REES46); the 'no' branch is exploratory



What we learned

What worked

- Warm start from a server pretrain: the biggest single gain
- On-device fine-tuning: small, significant gain, no upload
- INT8 and 8-bit uploads: within 0.15 points of full quality
- Pre-registration made the confirmatory answer credible
- Full-catalogue evaluation: comparable with published work

What did not

- A mis-scaled τ almost froze FL; the protocol missed it
- Federation over the 1.2M-user REES46 pretrain: no gain
- Device-only training from scratch: 1.64% of the cloud model
- Legacy DP randomness: no formal guarantee
- DP-FTRL and canary stopped; December replication not run (no data)

Conclusion and future work

Answer

- Warm-started federated edge system keeps **88.49%** of the cloud model
- Shipped pretrain alone 88.77%; on-device fine-tune adds +0.002283 MRR@20
- Exploratory: federation's value depended on server data and optimizer scale
- Fine-tune on device; federate only if server data is weak or unrepresentative
- Privacy must be engineered and audited, not assumed

Future work

- Distil a large teacher into the device model
- DP with secret randomness at full scale
- Real secure aggregation
- Replication on December 2019 data
- A field trial in the browser

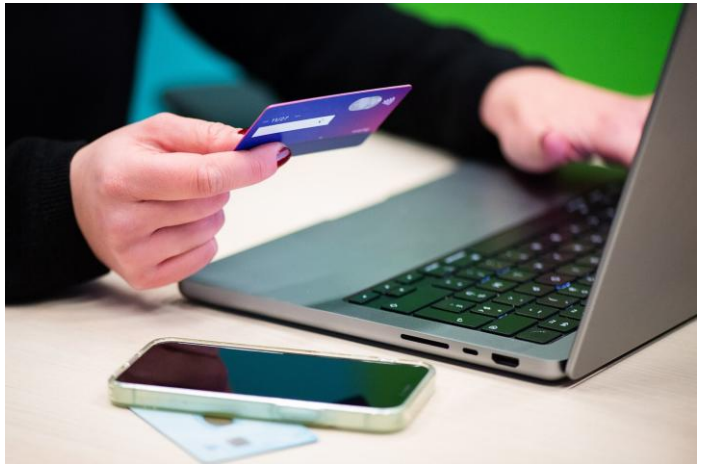


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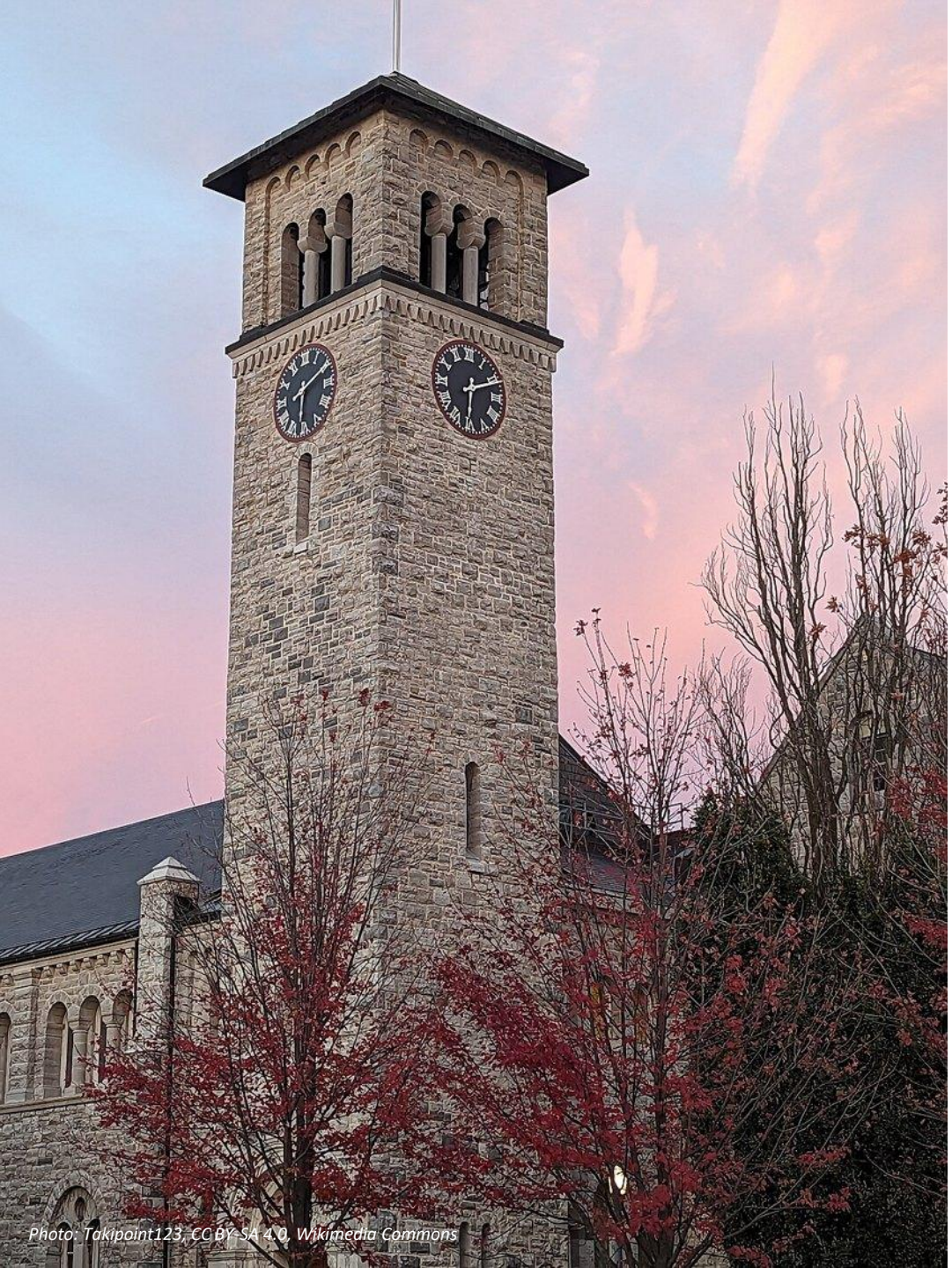


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Thank you

Questions and discussion

Code, results and checkpoints:
github.com/microsoftproject26/
Privacy-Preserving-Shopping-Intelligence